

```
from google.colab import files
uploaded = files.upload()
```

Choose Files netflix_titles.csv
netflix_titles.csv(text/csv) - 3399671 bytes, last modified: 9/30/2025 - 100% done
 Saving netflix_titles.csv to netflix_titles.csv

```
import pandas as pd
```

```
# Load dataset
df = pd.read_csv("netflix_titles.csv")
```

```
# Dataset Preview
print("First 5 rows of the dataset:")
```

First 5 rows of the dataset:

```
print(df.head())
```

```

  show_id  type      title      director \
0      s1  Movie  Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show      Blood & Water      NaN
2      s3  TV Show      Ganglands      Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans      NaN
4      s5  TV Show      Kota Factory      NaN

      cast      country \
0      NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalan...  South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...      NaN
3      NaN      NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...      India

  date_added  release_year  rating  duration \
0  September 25, 2021      2020  PG-13      90 min
1  September 24, 2021      2021  TV-MA  2 Seasons
2  September 24, 2021      2021  TV-MA  1 Season
3  September 24, 2021      2021  TV-MA  1 Season
4  September 24, 2021      2021  TV-MA  2 Seasons

      listed_in \
0      Documentaries
1  International TV Shows, TV Dramas, TV Mysteries
2  Crime TV Shows, International TV Shows, TV Act...
3      Docuseries, Reality TV
4  International TV Shows, Romantic TV Shows, TV ...

      description
0  As her father nears the end of his life, filmm...
1  After crossing paths at a party, a Cape Town t...
2  To protect his family from a powerful drug lor...
3  Feuds, flirtations and toilet talk go down amo...
4  In a city of coaching centers known to train I...
```

```
print("\nShape of the dataset:", df.shape) # rows, columns
```

Shape of the dataset: (8807, 12)

```
# Dataset Info
print("\nDataset Info:")
print(df.info())
```

```

Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   show_id      8807 non-null   object
1   type         8807 non-null   object
2   title        8807 non-null   object
3   director     6173 non-null   object
4   cast         7982 non-null   object
```

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5   country      7976 non-null   object
6   date_added   8797 non-null   object
7   release_year  8807 non-null   int64
8   rating       8803 non-null   object
9   duration     8804 non-null   object
10  listed_in    8807 non-null   object
11  description  8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
None

```

```

# Missing Values Check
print("\nMissing Values in Each Column:")
print(df.isnull().sum())

```

```

Missing Values in Each Column:
show_id      0
type         0
title        0
director    2634
cast         825
country      831
date_added   10
release_year  0
rating       4
duration     3
listed_in    0
description  0
dtype: int64

```

```

# Duplicate Check
print("\nNumber of Duplicates:", df.duplicated().sum())

```

```

Number of Duplicates: 0

```

```

# Basic Insights
print("\nUnique Content Types:", df['type'].unique())
print("Number of Unique Countries:", df['country'].nunique())
print("Number of Unique Genres:", df['listed_in'].nunique())

```

```

Unique Content Types: ['Movie' 'TV Show']
Number of Unique Countries: 748
Number of Unique Genres: 514

```

```

# Data Cleaning
# Fill missing country with 'Unknown'
df['country'].fillna('Unknown', inplace=True)

```

```

/tmp/ipython-input-1741672731.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained a
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df.co

    df['country'].fillna('Unknown', inplace=True)

```

```

# Drop duplicates
df.drop_duplicates(inplace=True)

```

```

# Descriptive Analysis
print("\nYear-wise Content Count (Release Year):")
print(df['release_year'].value_counts().sort_index())
print("\nTop 10 Countries with Most Content:")
print(df['country'].value_counts().head(10))
print("\nTop 10 Genres on Netflix:")
print(df['listed_in'].str.split(',').explode().value_counts().head(10))

```

```

Year-wise Content Count (Release Year):
release_year
1925         1
1942         2
1943         3

```

```

1944      3
1945      4
...
2017    1032
2018    1147
2019    1030
2020     953
2021     592
Name: count, Length: 74, dtype: int64

```

Top 10 Countries with Most Content:

```

country
United States    2818
India            972
Unknown          831
United Kingdom   419
Japan            245
South Korea      199
Canada           181
Spain            145
France           124
Mexico           110
Name: count, dtype: int64

```

Top 10 Genres on Netflix:

```

listed_in
International Movies    2624
Dramas                 1600
Comedies               1210
Action & Adventure      859
Documentaries          829
Dramas                 827
International TV Shows  774
Independent Movies      736
TV Dramas              696
Romantic Movies        613
Name: count, dtype: int64

```

Final Cleaned Dataset Info

```

print("\nCleaned Dataset Shape:", df.shape)
print("\n✅ Task 2 (EDA) Completed Successfully!")

```

Cleaned Dataset Shape: (8807, 12)

✅ Task 2 (EDA) Completed Successfully!